Disha Virmalwar

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Research Interests

Atmospheric teleconnections, tropical–extratropical interactions, and the circulation dynamics of extreme weather events.

Education

Indian Institute of Technology Kanpur **Aug 2018 – May 2023**
Integrated BS-MS Physics *Kanpur, India*

Technical Skills

Scientific Computing: Python (Xarray, Pandas, NumPy, Matplotlib), Climate Data Operators (CDO)
Data & Platforms: Jupyter, High-Performance Computing, NetCDF, Linux, LaTeX, Docker, GCP, SQL
Familiar: C/C++, MATLAB

Presentations

- "Revisiting 2025 Himalayan Extremes Through Tropical-Extratropical Interactions", ***Moist Convective Dynamics of Monsoon - II (Summer School)***, **ICTS-TIFR, Bangalore**, May 11-22, 2026 : *Poster Presentation*

Experience

Revisiting 2025 Himalayan Extremes through Tropical–Extratropical Interactions **Feb'26 - Present**
Prof. Akshaya Nikumbh, Prof. Vishal Dixit *(manuscript in preparation)* *CCS-IIT Bombay*

- Investigated extratropical precursors (blocking, Rossby wave breaking) and moisture-transport pathways (IVT, Low-Pressure Systems) associated with the 2025 Northwest India summer monsoon extremes.
- Built a 25-year atmospheric blocking and RWB catalog from ERA5/IMD data using Contrack and Wavebreaking packages on HPC, quantifying event intensity, frequency, and duration against climatology.
- Analyzed catastrophic 2025 summer monsoon precipitation events (EM-DAT based) to evaluate the Central Asian High (CAH) mediated precursor pathway, revealing distinct case-to-case mechanism variations.

Extratropical Influence on Subseasonal Variability and Predictability of Indian Summer Monsoon Rainfall (Mission Mausam III, MOES) **Aug'25 - Present**
Prof. Akshaya Nikumbh, Prof. V. Venugopal (IISC Bangalore) *IIT Bombay*

- Apply a Rossby wave-breaking detection framework to 40 years (1980-2019) of ERA5 potential vorticity data, computing Streamer and Cutoff indices to quantify wave-breaking events over the Indian region.
- Characterize the seasonal and latitudinal variability of these events and their relationship with ENSO phases.
- Implement wave-activity flux and blocking diagnostics to trace pathways of extratropical wave activity into the Indian region
- Develop a diagnostic package to quantify tropical–extratropical interactions and their relevance to monsoon variability and predictability.

Borrowing Base Portfolio Dashboard **Jun'23 - Jul'25**
FlairMinds Software Ltd *Pune, India*

- Architected scalable GCP data pipelines (processing 500k+ Parquet records) and developed automated scenario-analysis models, improving calculation accuracy by 25% and optimizing query accessibility for large datasets.
- Led a 5-person development team to build Looker dashboards; managed stakeholder alignment and delivered product pitches to streamline complex workflows.

Revisiting Butterfly Effect in Hydrodynamic Turbulence (Master's Project) **Jul'22 - Mar'23**
Prof. Mahendra Kumar Verma (Endterm Report) *IIT Kanpur, India*

- Studied sensitive dependence on initial conditions in Rayleigh–Bénard convection using 2D DNS (TARANG flow solver).
- Quantified perturbation growth using Lyapunov exponents and spectral energy analysis; found stronger growth at smaller scales.

Relevant Coursework

**** Audited***

- Applied Probability and Statistics
- Applied Numerical Methods
- Introduction to Climate Science *
- Ocean Dynamics *
- General Circulation of the Atmosphere and Oceans *

Position of Responsibility

Captain | *Institute Athletics Team, IIT Kanpur* **Mar'22 - Mar'23**

- Rebuilt the athletics team after a two-year gap, onboarding 60+ athletes, managing team finances, and coordinating 3 coaches across track, jump, and throw events.
- Led a 35-member squad in major sports meets, securing 1st overall at Udghosh'22 (out of 20+ teams) and 3rd place at InterIIT'22.